

SARASWATI VIDYA MANDIR
Half-Yearly Examination — Session 2026–27
SCIENCE (PHYSICS) — CLASS IX

Time Allowed: 1 Hour 30 Minutes

Maximum Marks: 23

General Instructions:

1. All questions are compulsory.
2. Marks for each question are indicated against it.
3. Draw neat diagrams wherever necessary.
4. Write the question number clearly before every answer.

SECTION A (*Very Short and Short Answer Questions*)

1. Define inertia. [1]
2. State Newton's third law of motion. [2]
3. (a) Write the SI unit of momentum. [1]
(b) A body of mass 5 kg is moving with a velocity of 4 m/s. Calculate its momentum. [2]
4. Distinguish between mass and weight. [2]
5. Why does a fielder lower his hands while catching a cricket ball? [2]

SECTION B (*Long Answer Questions*)

6. (i) State the law of conservation of momentum. [2]
(ii) Give one everyday example which illustrates this law. [1]
7. A force of 10 N acts on a body of mass 2 kg which is initially at rest. Find the acceleration produced and the velocity of the body after 3 seconds. [3]
8. What is meant by uniform circular motion? Give one example. [2]
9. Draw a velocity–time graph for a body moving with uniform acceleration, and explain what the slope of the graph represents. [3]
10. State any two applications of Newton’s first law of motion in daily life. [2]